

Rho Combinators as Sparse Linear Algebra

A classical route from reflective concurrency to data-parallel execution

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Abstract

The question this note answers is whether the concurrency of the rho calculus can be mapped onto the concurrency of a graphics processor by way of the name-free combinators, and specifically whether the reduction graph of a combinator term admits a matrix representation that factors into a representation of the rewrite rules and repeated application to a vector encoding of a term. The scope here is strictly classical: no weights, no amplitudes, no quantum reading.

The answer has three parts. First, the factorisation exists and is standard, once the right state space is identified. The companion logic note proves that the multiset of top-level atoms is a complete invariant for structural congruence, so a combinator state *is* an occupation vector and the one-step operator is a sum of ladder-operator monomials, one per rule, each of degree at most four. The equations of the calculus are quotiented by construction rather than by normalisation.

Second, the three invariants used in the draft’s expressiveness lattice — name closure, atom count, message count — turn out to be exactly the three finiteness conditions the matrix representation requires, and each is controlled by a different axis of the lattice. Name closure controls the index set, so the constructors decide whether the basis is finite or generated on demand. Atom count controls nilpotency, so the destructor decides whether the transfer operator is strictly block upper triangular. Message count controls occupation bounds. The lattice therefore already computes which matrix shape a term admits, and the eager and lazy parts of the operator are separated by the same analysis: seven of the twelve rules are constant-coefficient and fully schematic, four require an encode, and exactly one, the destructor, requires a decode.

Third, and this is the practical finding, the object worth multiplying is not the transition matrix of the reduction graph. It is the incidence matrix of atoms against names, which is term-sized and sparse. Redex finding for every rule is a join on that matrix, conflict detection between redexes is a second sparse product, and a machine step is a maximal independent set in the resulting conflict graph. Firing an independent set is sound with respect to the interleaving semantics, so the number of machine steps is the causal depth of the computation rather than its interleaving length. That ratio is the available speedup, and it is measurable. A reference implementation accompanies the note; its matrix formulation of redex finding agrees with naive enumeration on 400 random states, and on four families of terms the measured ratio of sequential steps to parallel steps is exactly the width of the term.

A phased development plan and seven numbered obligations close the note.

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1 What this note is

This note is a design analysis, not a completed theory. It records what the rho combinators of the draft in `F1R3FLY-io/publications/rho-combinators` already give a machine builder, what they do not, and in what order the remaining work should be attempted.

Scope. Classical execution only. Everything here is over the Boolean semiring or the natural numbers. The weighted, complex-valued and quantum readings developed elsewhere in the programme are deliberately absent; they attach to the same operator later, as §12 notes, and nothing below depends on them.

Claims. Six, and nothing more.

- (C1) A combinator state is an occupation vector over a species set, with structural congruence quotiented by construction (§4).
- (C2) The one-step relation is a sum of twelve ladder-operator monomials of degree at most four, schematic in the names (§5).

- (C3) The three invariants of the draft’s lattice section are the three finiteness conditions of that operator, one per lattice axis (§6).
- (C4) Upper triangularity holds exactly on the destructor-free fragment, and the general case needs either the history unfolding or nothing at all (§6.1).
- (C5) Redex finding and conflict detection are sparse matrix products on the atom-name incidence matrix, which is term-sized (§7).
- (C6) Firing a maximal independent set of redexes per step is sound with respect to the interleaving semantics, so machine steps count causal depth (§8).

Not claimed. That any of this beats a well-tuned sequential interpreter on real workloads; the measurements in §10 are on synthetic term families chosen to exhibit known widths, and the width of compiled `rho` programs is unmeasured. That the reduction graph itself should ever be materialised; the position taken here is the opposite. That maximal progress is the right resolver; it is *a* resolver, and Obligation 13.1 asks what it costs.

Status markers. **SETTLED** for what is proved or measured here or cited from the companion drafts; **ROUTE** for a construction believed to work whose details are not discharged; **OPEN** for a question with no proposed answer; **GATE** for something that must be settled before the work downstream of it is worth starting.

2 The two machines

The mapping asked for is between two concurrency paradigms, so it is worth writing down what each actually is before looking for a morphism.

2.1 What a graphics processor rewards

A modern accelerator is a bulk-synchronous machine with a very wide, very shallow execution model. It rewards, in rough order of importance:

- (G1) **Flat, indexable state.** Arrays of fixed-width records at predictable addresses. Pointer chasing through a heap of variably shaped nodes is the single most expensive thing one can ask of it.
- (G2) **Uniform work across lanes.** Threads in a warp that take the same branch. Data-dependent control flow serialises.
- (G3) **Bulk-synchronous phases.** A kernel launch does one thing to everything, then a barrier. Fine-grained locking and point-to-point synchronisation between work items are not available cheaply.
- (G4) **Standard sparse primitives.** Sparse matrix–vector and matrix–matrix products, segmented scans, sorts, and hash tables have mature, heavily optimised implementations. Work expressed in those terms inherits decades of tuning; work expressed otherwise does not.

2.2 What a tuple-space interpreter does

An RSpace-style interpreter for the ρ calculus does approximately the opposite of all four. Its state is a store of variably shaped terms reached by pointer; matching a receipt against a datum is a recursive walk over nested syntax with binder-sensitive pattern variables; conflicts between competing comm events are resolved by locking or by a single-threaded scheduler; and the inner loop is a hash lookup followed by unification, which is not a sparse linear algebra primitive by any reading.

None of that is a criticism of RSpace, which is a good design for a general machine. It is an observation that the gap between the two paradigms is concrete and itemised, and that a compiler standing between them has a specific job description.

2.3 The gap, and who closes it

The combinators close three of the four gaps by themselves, and the fourth is closed by the step semantics of §8.

Accelerator wants	What supplies it
(G1) flat indexable state	Combinator atoms are fixed-arity tuples of names; under presentation A every argument is a name, hence an integer index (§3).
(G2) uniform work	Twelve rules, each a fixed shape; one kernel per rule, no pattern matching at runtime. The pattern matching happened in the compiler.
(G3) bulk-synchronous phases	A step fires a maximal independent set of redexes, which is one barrier per causal layer (§8).
(G4) sparse primitives	Redex finding is a join, conflict detection is a product, independent-set selection is a standard parallel primitive (§7).

The compiler from the ρ calculus to the combinators is therefore load-bearing rather than decorative: it is the component that moves pattern matching from run time to compile time, and it costs only a constant factor in term size (draft 3, Prop. 8.1: the image has $O(|P|)$ atoms).

3 The combinators, and four structural facts

The calculus is that of draft 3, in the repaired form: two sorts, atoms

$$\mathbf{m}, \mathbf{d}, \mathbf{k}, \mathbf{fw}, \mathbf{b}_l, \mathbf{b}_r, \mathbf{s}, \mathbf{e}, \mathbf{q}, \mathbf{cons}_l, \mathbf{cons}_m, \mathbf{cons}_d, \mathbf{cons}_s,$$

the equations of an abelian monoid on $|$ together with $@*a = a$, and, under presentation A, $*@p = p$. The twelve rewrite rules are reproduced in Figure 1.

Four facts about this presentation do the work of the rest of the note. Three are theorems of the companion drafts; the fourth is a design recommendation.

(F1) **Atoms are flat.** Every atom is a head symbol applied to at most four arguments, and every argument is a name, except the payload of $\mathbf{q}$ and, under presentation B, of $\mathbf{m}$. There is no nesting, no binder, and no pattern variable.

(F2) **The multiset is complete.** By the logic note’s component lemma, $[P]$, the multiset of top-level atoms of the drop-free normal form, is a complete invariant: $P \equiv Q$ iff $[P] = [Q]$ as multisets. **SETTLED**

$d(a, b, c) \mid m(a, v) \longrightarrow m(b, v) \mid m(c, v)$	$b_l(a, b) \mid m(a, v) \longrightarrow fw(v, b)$
$k(a) \mid m(a, v) \longrightarrow \mathbf{0}$	$b_r(a, b) \mid m(a, v) \longrightarrow fw(b, v)$
$fw(a, b) \mid m(a, v) \longrightarrow m(b, v)$	$s(a, b, c) \mid m(a, v) \longrightarrow fw(b, c)$
$e(a) \mid m(a, v) \longrightarrow *v$	$fw(a, b) \mid q(a, p) \longrightarrow m(b, @p)$
$cons_l(a, b, c) \mid m(a, @p) \mid m(b, @r) \longrightarrow m(c, @(p \mid r))$	
$cons_m(a, b, c) \mid m(a, u) \mid m(b, v) \longrightarrow m(c, @m(u, v))$	
$cons_d(a, b, c, f) \mid m(a, u) \mid m(b, v) \mid m(c, w) \longrightarrow m(f, @d(u, v, w))$	
$cons_s(a, b, c, f) \mid m(a, u) \mid m(b, v) \mid m(c, w) \longrightarrow m(f, @s(u, v, w))$	

Figure 1: The twelve rules, closed under $\mid$ and under the equations, with subjects matched up to $\equiv_{\mathcal{N}}$.

(F3) **Rules are small and subject-joined.** Every rule has between two and four premises, each premise a single atom, and the premises share names. The logic note remarks that this is the only structural fact its modality uses; it is also the only structural fact the machine below uses. **SETTLED**

(F4) **Name equality is decidable and internable.** On closed drop-free normal forms $\equiv_{\mathcal{N}}$ is decidable, by normalisation followed by multiset comparison, and $@-$ is injective there. So names can be hash-consed to integers once and compared by integer equality thereafter. **SETTLED**

Remark 3.1 (Presentation A is the machine presentation). Draft 3 leaves the choice between presentations A and B open on grounds of where decoding happens. For a machine the choice is forced, and in favour of A. Under A a message payload is a name, hence an integer, so every atom is a fixed-width tuple of integers and (G1) is satisfied exactly. Under B a payload is a process, so an atom is a tuple of integers together with a variable-sized block, and the store becomes ragged.

The same move should be made for q . Although its payload is sorted as a process, storing it by its quote — holding $q(a, p)$ as the pair $(a, @p)$ — loses nothing, because $@-$ is injective on normal forms by (F4), and it makes the **quote** rule a field copy rather than a construction. This is the difference between rule class (a) and class (b) in §5, and it is obtained for free. **SETTLED**

4 States are occupation vectors

Definition 4.1 (Species). A *species* is a pair of a head symbol and a tuple of names of the matching arity: the set is $\mathcal{S} = \{(A, \vec{n}) : A \text{ a head symbol}, \vec{n} \in \mathcal{N}^{\text{ar}(A)}\}$, where $\mathcal{N}$ is the set of names modulo $\equiv_{\mathcal{N}}$.

By (F2) a state is a finite multiset over $\mathcal{S}$, that is, a function $\gamma : \mathcal{S} \rightarrow \mathbb{N}$ of finite support. Write $\mathbb{F}$ for the free module on the set of such γ and $|\gamma\rangle$ for the basis vector at γ . Equivalently, and more usefully, $\mathbb{F}$ is the free commutative algebra generated by $\mathcal{S}$: a state is a monomial $\prod_{\sigma} (a_{\sigma}^{\dagger})^{\gamma(\sigma)} |\mathbf{0}\rangle$, and the abelian monoid equations of the calculus are the commutativity of that product.

Principle 4.2 (Structural congruence is free). The equations of Definition 3.2 of draft 3 are not imposed on the representation and are not discharged by a normalisation pass. Commutativity, associativity and the unit of $\mid$ are the algebra of the multiset; $*@p = p$ and $@*a = a$ are discharged once, at internig time, by the drop-elimination lemma. No run-time cost. **SETTLED**

This is the first place where the combinators pay. In a direct **rho** machine, matching modulo structural congruence is a recurring cost and a recurring source of subtlety; here it is a property of the data structure.

4.1 Names, and the self-indexing problem

$\mathcal{S}$ is indexed by $\mathcal{N}$, and $\mathcal{N}$ is the set of quoted processes, so by (F2) it is the set of finite multisets over $\mathcal{S}$. The index set of the basis therefore satisfies

$$\mathcal{N} \cong M_{\text{fin}}(\mathcal{S}), \quad \mathcal{S} = \coprod_A \mathcal{N}^{\text{ar}(A)}.$$

The basis of the state space indexes itself. This is reflection, seen through the linear algebra, and it is the reason no finite matrix can be written down in advance for the full calculus. The solution of the domain equation is the initial algebra: names are finite trees, so $\mathcal{N}$ is countable, finitely branching, and graded by depth of quotation.

Definition 4.3 (The intern table). The machine realises $\mathcal{N}$ as a bijection between integers and canonical component multisets, in two directions:

$$\begin{aligned} \text{quote} : M_{\text{fin}}(\mathcal{S}) &\rightarrow \mathbb{N} && \text{hash-cons; an encode,} \\ \text{drop} : \mathbb{N} &\rightarrow M_{\text{fin}}(\mathcal{S}) && \text{table read; a decode.} \end{aligned}$$

Remark 4.4 (Interning order is a gauge freedom). Two runs that fire the same rules in different orders may assign different integers to the same name, because integers are handed out on insertion. The resulting states are equal up to the canonical bijection determined by **quote**, which is injective. Nothing observable depends on the assignment, and the machine must never let an integer comparison stand in for a semantic distinction between runs. **SETTLED**

5 The rule operator

Definition 5.1 (Ladder operators). For $\sigma \in \mathcal{S}$, let $a_\sigma^\dagger$ add one occurrence of σ and a_σ remove one, with $a_\sigma|\gamma\rangle = 0$ when $\gamma(\sigma) = 0$.

Definition 5.2 (The transfer operator). For a rule r with premise multiset L_r and conclusion multiset R_r , both over atom shapes with name variables, and for a substitution θ of those variables by names,

$$H_r = \sum_{\theta} \left(\prod_{\beta \in R_r \theta} a_\beta^\dagger \right) \left(\prod_{\alpha \in L_r \theta} a_\alpha \right), \quad T = \sum_r H_r.$$

This is the factorisation asked for. T is a representation of the rewrite rules alone: it is schematic in the names, independent of any particular term, and of degree at most four by (F3). The unfolding of the reduction graph is repeated application of T to $|\gamma\rangle$.

Proposition 5.3 (Matrix elements count derivations). $\langle \gamma' | H_r | \gamma \rangle$ is the number of distinct occurrence tuples in γ matching L_r and producing γ' , namely $\prod_{\alpha} (\gamma(\alpha))_{\underline{m}_\alpha} / \prod_{\alpha} m_\alpha!$ where m_α is the multiplicity of α in $L_r \theta$ and $(n)_{\underline{m}}$ is the falling factorial.

Proof. Standard for multiset rewriting: a match is a choice of m_α distinct occurrences of each premise species, unordered within a species. $\square$

Remark 5.4 (This is mass action, and it is not an accident). Proposition 5.3 is the mass-action law, and the coefficient is the same multiplicity h that the Gillespie implementation in **rho-life** already computes. The combinators are a chemical reaction network with twelve reaction schemes. The classical reading taken here uses only the Boolean shadow of that coefficient; the weighted readings elsewhere in the programme use the coefficient itself.

5.1 Three classes of rule

The twelve rules are not alike in how much of T can be written down ahead of time.

Class (a), seven rules: permutation of existing indices.

`dist`, `kill`, `forward`, `bindl`, `bindr`, `synch`, `quote`. The conclusion is a fixed multiset of atoms whose names all occur in the premise. H_r is a constant-coefficient ladder monomial, fully schematic, requiring no access to the intern table at all. Note that `quote` belongs here only because of Remark 3.1.

Class (b), four rules: encode.

The four constructors. The conclusion is a single atom, so the arity is fixed, but its payload is a *new* index, obtained by hash-consing a term built from the premise names. H_r grows the index set.

Class (c), one rule: decode.

`eval`. It is the only rule whose conclusion multiset has a size depending on the *value* of a premise name. Its rows cannot be written schematically; they require reading the intern table.

Principle 5.5 (Where the laziness is, and which kind). The matrix must be materialised lazily for two distinct reasons, and the lattice separates them. Class (b) makes the *basis* generate on demand: new columns appear. Class (c) makes the *rows* generate on demand: an existing column's image is not known until the index is decoded. Class (a), which is most of the calculus, is fully eager. SETTLED

6 Three invariants, three finiteness conditions

Section 9 of draft 3 proves three invariants in order to draw the expressiveness lattice. They are, without modification, the three finiteness conditions a matrix representation needs, and each is governed by a different axis of the lattice. This is the note's main structural observation.

Theorem 6.1 (The lattice classifies the matrix). *Let T be a combinator term over a constructor set S .*

- (i) (Index set; constructor axis.) *By the name-closure lemma, every name occurring in a reduct of T lies in the closure of $\mathcal{N}^*(T)$ under the node shapes $\{A : \text{cons}_A \in S\}$. If $S = \emptyset$ the reachable name set is $\mathcal{N}^*(T)$, which is finite, so $\mathcal{S}$ restricted to reachable species is finite of size $O(|\mathcal{N}^*(T)|^4)$. If $S \neq \emptyset$ the reachable name set is the free S -algebra on $\mathcal{N}^*(T)$, countable and graded by depth.*
- (ii) (Nilpotency; destructor axis.) *By the atom-count lemma every rule but `eval` strictly decreases the number of non-message atoms. Let N be that number in T . Without `e`, the function $\gamma \mapsto N(\gamma)$ is a grading strictly decreased by every step, so the reachable sub-operator is strictly block upper triangular with respect to it and $T^{N+1} = 0$.*
- (iii) (Occupation bound; duplication axis.) *By the message-count lemma `d` is the only rule that increases the number of messages without consuming a `q` or opening a payload, so without `d` and without `e` the message count is non-increasing and occupation numbers are bounded by those of T .*

Proof. Each clause is the cited lemma read as a statement about the operator. For (ii), a strictly decreasing integer grading bounded below by 0 and above by N gives a block decomposition in which T maps block j into blocks $< j$, which is the definition of strict block upper triangularity, and composition of $N + 1$ such maps is zero. $\square$

Corollary 6.2 (Closed-form reachability on the destructor-free fragment). *Without $\mathbf{e}$, the reachability closure is the terminating Neumann series $\sum_{k \leq N} T^k = (\mathbf{1} - T)^{-1}$, computable in $\lceil \log_2 N \rceil$ repeated squarings, each a sparse matrix product. **SETTLED***

Remark 6.3 (One atom, three characterisations). The destructor $\mathbf{e}$ is, simultaneously: the carrier of Turing power, by the draft’s strong-normalisation proposition; the unique breaker of the grading, by Theorem 6.1(ii); and the unique rule whose operator cannot be written schematically, by §5.1. Three independent characterisations of the same atom, arrived at from logic, from order theory and from implementation. The draft’s remark that “the destructor carries the computational power that reflection supplies” acquires an operational reading: the power is exactly the ability to turn an index into a state.

Remark 6.4 (The Petri net reading). Without constructors and without $\mathbf{e}$, the combinators are a place/transition net: places are species, transitions are rule instances, and the step semantics of §8 is the net’s step semantics. Adding constructors makes the place set generated rather than given; adding $\mathbf{e}$ makes the transition relation depend on the contents of a place. The lattice is therefore also a descent from nets through generated-place nets to Turing completeness, which is a second reading of the same three invariants and may be worth a paragraph in the draft. **ROUTE**

6.1 Upper triangularity, precisely

The proposal that opened this line of work was to encode the reduction graph as an upper triangular matrix. Upper triangularity is equivalent to acyclicity of the graph, and the reduction graph of a combinator term is not acyclic in general: reflection admits terms that step to a structurally congruent copy of themselves, and by Principle 4.2 structurally congruent means equal. So on the nose the encoding is available exactly on the destructor-free fragment, by Theorem 6.1(ii).

Two repairs are available, and for the classical execution goal the second is recommended.

- (1) **Work in the unfolding.** The free history of a term is the universal cover of its reduction graph, graded by step count, hence acyclic by construction; triangularity is then free, at the cost of never identifying a returning state. The cycles one gives up are exactly the non-trivial elements of π_1 that the history endofunctor’s erasure lattice already parametrises, so the apparatus exists. This is the right route if the closure is the object wanted, for instance for model checking. **ROUTE**
- (2) **Do not ask for it.** For running a program one needs a frontier, not a closure: the current state and its successors. Triangularity is a property of the whole graph and buys nothing at the frontier. A general sparse non-negative operator applied to a sparse vector is entirely adequate, and iterative methods are indifferent to cycles. **SETTLED**

7 What actually gets multiplied

There are three matrices in play and conflating them is the easiest mistake to make here.

	Indexed by	Size, and what it is for
T	markings $\times$ markings	Astronomically large, never materialised. The semantics.
B	redexes $\times$ occurrences	$O(\text{redexes})$ nonzeros. Conflict detection.
A	atom occurrences $\times$ names	Term-sized , exactly $\sum_{\text{atoms}} \text{ar}(A)$ nonzeros. The thing the machine multiplies.

Definition 7.1 (Incidence). For each head symbol A and each argument slot $j < \text{ar}(A)$, let M_j^A be the Boolean matrix with $M_j^A[i, n] = 1$ iff the i -th occurrence of A carries name n in slot j . Each row has exactly one nonzero; the whole family has $\sum_{\text{atoms}} \text{ar}(A) \leq 4|\gamma|$ nonzeros.

7.1 Redex finding is a join

By (F3) every rule is a conjunction of atom premises sharing names, which is a relational join on the incidence matrices. Writing $\cdot$ for Boolean sparse matrix product:

Rules	Expression	Cost
the seven two-premise rules against m quote	$M_0^C \cdot (M_0^m)^\top$	one SPGEMM; nonzeros $\sum_n c_n \mu_n$ where c_n, μ_n are the numbers of consumers and messages at n
$\text{cons}_l, \text{cons}_m$	$M_0^{\text{fw}} \cdot (M_0^q)^\top$	as above
	$X = M_0^{\text{cons}} \cdot (M_0^m)^\top$, $Y = M_1^{\text{cons}} \cdot (M_0^m)^\top$, then the row-wise Cartesian product	two SPGEMMs plus a segmented expansion
$\text{cons}_d, \text{cons}_s$	$X \bowtie Y$ with $j \neq k$ masked out three such products, then a three-fold row-wise product with distinctness masking	three SPGEMMs plus expansion

Remark 7.2 (Contention is a join cost, not a scheduling cost). The two-premise product has $\sum_n c_n \mu_n$ nonzeros, which is quadratic in the occupancy of a contended channel. This is real and is measured in §10 (experiment E2, where c consumers and c messages on one channel produce c^2 redexes). It does *not* cost parallelism: the maximal independent set on that quadratic redex set is a perfect matching of size c , found in one step. The cost is enumeration, and the standard remedy — worst-case-optimal join, or bucketing by name before the product — applies unchanged. OPEN which remedy is right at scale.

7.2 Conflict detection is a second product

Definition 7.3 (Conflict). Two redexes conflict if they consume a common occurrence. With B the redex-occurrence incidence matrix, the conflict graph is the off-diagonal nonzero pattern of $B \cdot B^\top$.

So the entire per-step analysis — which redexes exist, and which of them may fire together — is two rounds of Boolean sparse matrix multiplication on term-sized operands. That is the concrete content of “rho execution as linear algebra” in the classical setting, and it involves no matrix whose dimension is the number of states.

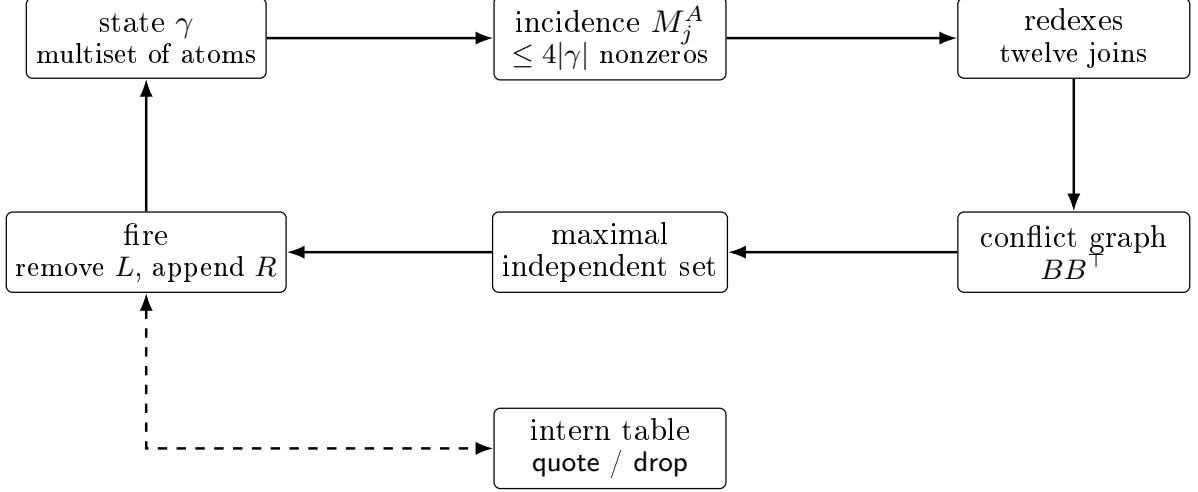


Figure 2: One machine step. Every solid arrow is a data-parallel bulk operation. The dashed arrow is the encode and the decode of §9, used only by rule classes (b) and (c).

8 The parallel step

This section supplies (G3), and is where the concurrency of the calculus meets the concurrency of the machine.

Definition 8.1 (Independence, step). Redexes r_1, r_2 enabled in γ are *independent* if the occurrence multisets they consume are disjoint. A set S of pairwise independent redexes is a *step*; $\gamma \Rightarrow_S \gamma'$ removes every occurrence consumed by a member of S and adds every atom produced by a member of S .

Lemma 8.2 (Diamond). *If r_1, r_2 are independent and enabled in γ , then each remains enabled after the other fires, and both orders yield the same state.*

Proof. Firing removes only the occurrences consumed, so by disjointness the premises of r_1 survive the firing of r_2 and conversely. The produced multiset of r_i is a function of the occurrences r_i consumes and of the intern table; by Remark 4.4 the table's contribution is order-independent up to the canonical bijection. Both orders therefore give $\gamma - L_{r_1} - L_{r_2} + R_{r_1} + R_{r_2}$. $\square$

Theorem 8.3 (Step soundness). *If $\gamma \Rightarrow_S \gamma'$ then $\gamma \longrightarrow^* \gamma'$ by a sequential derivation firing exactly the members of S , in any order, and all such orders agree.*

Proof. Induction on $|S|$ using Lemma 8.2. $\square$

Corollary 8.4 (Steps count causal depth). *A parallel run of k steps performs $\sum_i |S_i|$ rewrites. The available speedup over a sequential interpreter is therefore the mean step width, and the number of machine steps is bounded below by the causal depth of the derivation.*

Remark 8.5 (Maximal progress is a resolver, not a theorem). The converse of Theorem 8.3 fails, and deliberately. Requiring S maximal excludes interleavings in which an enabled, unconflicted redex is left unfired; and when redexes do conflict, the choice of which enters S is precisely the choice a scheduler makes. In the vocabulary of the programme's semantics work, maximal progress is a *resolver* for the calculus's non-determinism, of the same kind as physical core races in an RSpace implementation, and not the same resolver. It is also the true-concurrency reading: a maximal-progress machine observes a partial order of causes rather than a sequence, which is what the programme has elsewhere argued the rewrites should determine. See Obligation 13.1.

Remark 8.6 (Contention sets, generalised). The conflict structure is a hypergraph, since rules have up to four premises, so the object here is an independent set in the conflict graph of redexes rather than a matching in a bipartite graph. This is the k -ary generalisation of the contention-set normalisation and maximal matchings developed for the graded where-clause work, and the two should be reconciled before either is implemented twice. **GATE**

Remark 8.7 (Selection is a standard primitive). A maximal independent set with random priorities is a classical parallel algorithm running in $O(\log n)$ expected rounds, and re-randomising the priorities each step makes starvation of any given redex have probability tending to zero. The reference implementation uses the naive greedy version; a device implementation should use the standard one.

9 Reflection, at the level of the machine

Rule classes (b) and (c) are the two lazy parts of the operator. It is worth saying plainly what they cost, because the answer is surprising.

Decode, the eval rule.

$e(a) \mid m(a, v) \longrightarrow *v$ reads the intern table at v and appends the stored component multiset to the state. On a device this is a table lookup followed by a bulk copy of a contiguous block: among the cheapest things an accelerator does.

Encode, the constructors.

$\text{cons}_m(a, b, c) \mid m(a, u) \mid m(b, v) \longrightarrow m(c, @m(u, v))$ builds a small canonical term and hash-conses it. Concurrent hash tables with insert-or-get semantics are a standard device primitive.

Principle 9.1 (Reflection is an intern table). The feature that makes the calculus theoretically hard — that a name is a program and a program can be recovered from a name — is, at the level of the machine, exactly a bidirectional intern table. The theoretical difficulty and the implementation difficulty do not coincide here, and they point in opposite directions: reflection is what obstructs a static matrix and what makes the dynamic one cheap, because storing a continuation as an index is what keeps atoms fixed-width.

This also explains the succinctness result of draft 3 from the machine side. Yoshida’s elimination must rewrite a continuation to delay it, costing $\Theta(3^d)$ in nesting depth; the reflective encoding stores it as a name and pays for a gate. A machine that stores continuations as integers in a table is doing the same thing, and the $O(|P|)$ bound on the encoding is what keeps the state array small enough to be worth the transfer to a device.

Remark 9.2 (Two serialisation points). The intern table is shared mutable state and is therefore the one place the bulk-synchronous discipline is violated. Both operations are commutative and idempotent in effect, by Remark 4.4, so the table can be updated by an unordered batch insert per step rather than by locking. Whether the table should also be garbage collected, and on what criterion, is Obligation 13.3.

10 Measurements

A reference implementation accompanies this note (`rhocombmat.py`, `experiments.py`). It represents a state as a multiset of atom occurrences, builds the incidence matrices of Definition 7.1 with `scipy.sparse`, finds redexes by the products tabulated in §7, builds the conflict graph as $BB^\top$, selects a maximal independent set, and fires it. It is a correctness reference, not a performance artefact; the numbers below are step counts and widths, not timings.

Empirical finding 10.1 (The join formulation is correct). On 400 randomly generated states over all thirteen head symbols and two to five distinct names, the redex set computed by sparse matrix products agrees exactly with independent naive enumeration: 400/400.

Empirical finding 10.2 (Steps count width). On four term families, averaged over five seeds:

Term	atoms	par. steps	rewrites	seq. steps	max width	ratio
E1 fan, $k = 1, d = 8$	9	8.00	8	8	1	1.00
E1 fan, $k = 4, d = 8$	36	8.00	32	32	4	4.00
E1 fan, $k = 16, d = 8$	144	8.00	128	128	16	16.00
E1 fan, $k = 64, d = 8$	576	8.00	512	512	64	64.00
E2 contended, $c = 4$	8	1.00	4	4	4	4.00
E2 contended, $c = 16$	32	1.00	16	16	16	16.00
E3 duplicator tree, $h = 2$	8	3.00	7	7	4	2.33
E3 duplicator tree, $h = 4$	32	5.00	31	31	16	6.20
E3 duplicator tree, $h = 6$	128	7.00	127	127	64	18.14
E4 construct + eval, $k = 1$	4	2.00	2	2	1	1.00
E4 construct + eval, $k = 8$	32	2.00	16	16	8	8.00
E4 construct + eval, $k = 32$	128	2.00	64	64	32	32.00

E1 is k independent forwarder chains of length d : parallel steps are exactly d , independently of k , and the ratio is exactly k . E2 is c messages and c consumers on a single channel: one parallel step, at the cost of c^2 enumerated redexes (Remark 7.2). E3 is a binary duplicator tree of height h : parallel steps are $h + 1$ against $2^{h+1} - 1$ sequential steps. E4 exercises rule classes (b) and (c) together, building a name with `consm` and opening it with `e`.

Remark 10.3 (What these do and do not establish). They establish that the construction is correct and that the step count is the causal depth on families whose depth is known analytically. They establish nothing about real `rho` programs, because the width of the combinator image of a compiled `rho` term has not been measured. That measurement is the first item in §11 for a reason: it is the gate on the whole enterprise, and it requires the compiler, not the machine. GATE

Principle 10.4 (The width thesis). The speedup available from this route is the mean width of the reduction, and nothing else. Sequential terms gain nothing; the machine adds a constant per-step overhead of two sparse products and an independent-set selection. The thesis is falsifiable by measuring the width distribution of compiled `rho` programs, and should be falsified or confirmed before any device work is undertaken.

11 Recommendations

Seven phases, each with an exit criterion. Phases P0–P2 are host-side and establish whether the enterprise is worth pursuing; P3 onward are the device work and should not begin until P2’s number is in hand.

P0. Presentation and storage decisions.

Adopt presentation A; store the payload of `q` by its quote (Remark 3.1). Build the intern table with canonical multiset hashing of names. *Exit*: $\equiv_{\mathcal{N}}$ decided by integer comparison on a corpus of terms, cross-checked against structural comparison.

P1. The compiler, instrumented.

Complete the `rho`-to-combinators compiler already scoped in the `rho-combinators` programme, and instrument it to emit the atom multiset directly in the machine’s represen-

tation. *Exit*: the $O(|P|)$ bound observed on a corpus; round-trip against a ρ interpreter on a behavioural test suite.

P2. The width measurement.

Run the host reference machine on the compiled images of real programs and record the width distribution per step. *Exit*: a number. Principle 10.4 stands or falls here. **GATE**

P3. Host implementation in earnest.

Port the reference machine to Rust against a sparse Boolean backend, one kernel per rule, with bucketing by name ahead of the two-premise products. *Exit*: differential agreement with the sequential interpreter on reachable-state sets for the destructor-free fragment, where Corollary 6.2 gives an independent closed form.

P4. The device port.

Move the incidence matrices, the twelve join kernels, the conflict product and the independent-set selection to the accelerator; keep the intern table host-side initially, then move it behind a batched insert-or-get. *Exit*: identical step traces to P3 under a fixed priority seed.

P5. Contention remedies.

Replace the naive two-premise product with a worst-case-optimal join or name-bucketed product where profiling shows $\sum_n c_n \mu_n$ dominating. *Exit*: enumerated redexes per step within a constant factor of fired redexes on the P2 corpus.

P6. The by-product for the logic.

On the destructor-free fragment, expose $(\mathbf{1} - T)^{-1}$ by repeated squaring, and use it as the model checking engine for the spatial-behavioural logic of the companion note, where the modality is a matrix-vector product and the fixed-point operators are iterations of it. *Exit*: agreement with a direct model checker on the lattice-occupancy formulae.

Remark 11.1 (Why P6 is not an afterthought). The transfer operator's natural consumer is the logic, not the interpreter. The modality $\langle K \rangle \varphi$ is a sum over successors, which is $T^\top$ applied to the characteristic vector of φ , and the $\mu X(\vec{a}).\varphi$ operator is an iteration of that. The companion note's requirement that the lattice-occupancy formulae be decidable and Theorem 6.1's nilpotency condition are plausibly the same condition seen twice; establishing that would be a genuine result and costs little once P3 exists. **ROUTE**

12 What this does not do

Stated plainly, so that no one is disappointed later.

- **No asymptotic improvement to a single sequential run.** The total number of rewrites is unchanged. Only their scheduling changes.
- **No help for narrow terms.** A term whose reduction is a chain gains nothing and pays the per-step overhead.
- **No reachability decision procedure.** Reachability remains undecidable in the presence of $\mathbf{e}$; the operator is locally finite, so finite iteration is always well defined, but the closure need not be reached.
- **No treatment of weights.** Everything here is Boolean. The weighted reading attaches at exactly one place — a diagonal factor D_r in H_r whose entries depend on the premise names — and the combinators, having no where-clause, would need the weights supplied per rule instance rather than per spatial formula. That is a separate note.

- **No claim about per-copy identity.** Because $\equiv$ is multiset equality, two occurrences of the same atom are genuinely indistinguishable. Machine-level occurrence identities exist only to define conflict and must not leak into the semantics. Any downstream construction that wants per-derivation identity has to get it from somewhere else.

13 Obligations

Obligation 13.1 (Resolver fidelity). Maximal progress with re-randomised priorities induces a distribution over runs. Characterise it, and say how it differs from the distribution induced by uniform choice of a single redex. The question matters because the programme elsewhere treats the resolver as a parameter of the interpretation, and a machine that silently fixes one has made a semantic commitment. **OPEN**

Obligation 13.2 (Canonical name hashing). $\equiv_{\mathcal{N}}$ is decidable by recursive multiset comparison, but the machine wants a canonical hash computable bottom-up in one pass over the name’s tree, stable across runs. Give one, and bound its collision behaviour on the corpus. **ROUTE**

Obligation 13.3 (Garbage). Inert atoms accumulate: a k with no message, a message with no consumer, a q nobody quotes. They cost incidence nonzeros on every step. Decide whether inertness is detectable cheaply — a species with no possible partner under any rule is detectable by a name-degree test — and whether the intern table needs collection at all. **OPEN**

Obligation 13.4 (Contention set reconciliation). Reconcile the independent-set construction of §8 with the contention-set normalisation of the graded where-clause work, which solves the binary case. One of the two should be derived from the other. **GATE**

Obligation 13.5 (Occupancy decidability and nilpotency). Determine whether the companion logic note’s requirement that lattice-occupancy formulae be decidable coincides with the finiteness conditions of Theorem 6.1. **ROUTE**

Obligation 13.6 (Fairness). Prove or refute that re-randomised-priority maximal progress is fair, in the sense that a continuously enabled redex fires with probability one. **OPEN**

Obligation 13.7 (Width of real programs). Measure the width distribution of the combinator images of compiled rho programs. This is P2 and it gates everything downstream. **GATE**

14 Conclusion

The proposal that opened this line of work — encode the reduction graph as an upper triangular matrix, factored into a representation of the rules and repeated application to a vector — survives with one correction and one redirection.

The correction is that triangularity is available exactly on the destructor-free fragment, where the atom-count grading supplies it, and that in general one either moves to the history unfolding or does without. The redirection is that the matrix worth building is not the one indexed by states. It is the incidence of atoms against names, which is the size of the term; on it, redex finding and conflict detection are two sparse Boolean products, and a machine step is a maximal independent set. The factorisation asked for does exist, in the form of twelve ladder-operator monomials, and it is the right way to *think* about the semantics; but the thing to *multiply* is smaller and nearer to hand.

What makes this work at all is the combinators themselves. Because they are name-free and flat, a state is a multiset, structural congruence is free, pattern matching moves into the compiler, and every atom becomes a fixed-width tuple of integers. Those four properties are the ones an accelerator asks for, and the calculus has them by construction rather than

by engineering. The remaining risk is not in the representation. It is whether compiled rho programs are wide, and that is a measurement, not a theorem.

Acknowledgement. This note was drafted in dialogue with Claude (Anthropic), which also wrote and ran the reference implementation.

A The reference implementation

Two files accompany the note. `rhocombmat.py` is the machine: an intern table, a state as a multiset of occurrences, incidence matrices, the twelve join expressions, the conflict product, a greedy maximal independent set, and the firing rules including the encode and decode paths. `experiments.py` builds the four term families, validates the matrix join against naive enumeration, and produces the table of Finding 10.2. The core of the step is short enough to quote:

```
def run_parallel(state, nt, seed=0, cap=10000):
    """Bulk-synchronous: each step fires a maximal independent set."""
    rng = random.Random(seed)
    steps, fired, widths, counts = 0, 0, [], []
    while steps < cap:
        R = redexes_matrix(state, nt) # twelve joins on the incidence
        if not R:
            break
        C, nnz = conflict_graph(R, state) # B B^T
        sel = mis(R, C, rng) # maximal independent set
        widths.append(len(sel))
        counts.append(len(R))
        for r in sel:
            fire(state, nt, R[r]) # remove L, append R
        fired += len(sel)
        steps += 1
    return steps, fired, widths, counts
```

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